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Research papers

Optimal participation of a wind and hybrid battery storage system in the day-ahead and automatic frequency restoration reserve markets

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ABSTRACT

To evaluate the potential market revenue increase coming from the installation of a hybrid battery energy storage system (HESS) paired with a wind plant, a model is proposed for optimizing the simultaneous participation of the plant in the day-ahead (DA) and the automatic frequency restoration reserve (aFRR) markets. In the analysis, the Danish market is assumed as a case study. The HESS is composed of a combination of lithium-ion (LiB) and vanadium redox flow batteries (VRFB). The optimal dispatching problem is solved as a Mixed Integer Linear Program (MILP), using a robust formulation to consider the uncertainty of the aFRR market. The developed economic model incorporates the influence of cyclic degradation on the lithium-ion battery's life, considering it dependent on the number of cycles and the depth-of-discharge of each cycle. The economic and technical effects of hybridization are investigated for different storage sizes. Furthermore, the impact of robust optimization, in contrast to non-robust optimization, is investigated along with the impact of detailed storage models, which consider variable efficiencies and variable charging and discharging rates, in contrast to simple models, which consider constant storage efficiencies. Results show that the hybridization between lithium-ion and vanadium batteries positively influences the life of lithium-ion batteries, increasing its life by up to 47% compared to a case without hybridization. For a 10 MW wind farm, the optimal battery configuration is comprised of a hybrid system made of a 1 MW/1MWh LiB and a VRFB of negligible size, with 2 years of payback time. The simultaneous participation in both the DA and aFRR markets proves to be advantageous with annual revenue increases of 39%–56% compared to the system bidding on the sole DA market, regardless of the storage size. The revenue increase attributed to the battery installation does not ensure the feasibility of investment, due to the high cost of the battery technologies, even if the hybrid configuration proves to be more profitable than the non-hybrid configuration.

1. Introduction

The integration of variable renewable energy sources, such as wind and solar power, into the grid, presents challenges due to the variability and intermittency of these sources. Energy storage systems can play a significant role in addressing these challenges by providing a means to store excess energy during periods of high generation and release it during periods of low generation [1]. It is estimated that with a 20%–30% share of intermittent RES in the grid, there is still enough conventional generation to provide the required stability, for most of the time. When the renewable penetration approaches shares of 50%, instability could happen more frequently, and new solutions for integration and system stability will be needed. Differently, from an energy balance point of view, previous researches suggest that storage systems will only become necessary if RES share reaches around 80% [2]. In 2020, the two leading European countries in terms of share of electricity generation from variable renewable energy had a penetration of 36% (Ireland) and 63% (Denmark) [3], while the share of total renewable energy sources in gross electricity consumption in the EU was 37.4% [4]. The renewable energy integration in EU's power sector is projected to reach over 70% of gross electricity generation by 2030 [5]: this development, mainly driven by the EU's decarbonization targets [6], will make it essential to rethink the structure of the electricity markets, increasing the grid flexibility soon [5].

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Table 1
List of abbreviations.

Abbreviation	Description
aFRR	Automatic frequency restoration reserve
BESS	Battery energy storage system
DA	Day-ahead
DOD	Depth-of-discharge
DPBP	Discounted payback period
FCR	Frequency containment reserve
HESS	Hybrid energy storage system
LiB	Lithium-ion battery
MILP	Mixed integer linear program
mFRR	Manual frequency restoration reserve
NPV	Net present value
RES	Renewable energy source
RO	Robust optimization
SOC	State of charge
SOE	State of energy
TSO	Transmission system operator
VRFB	Vanadium Redox Flow Battery

While the concept of utilizing storage systems to facilitate the integration of renewable energy into the grid has been a subject of discussion within the smart grid community for a long time, there remains ambiguity regarding how to stimulate substantial investments in the construction of large, privately owned storage facilities and how to effectively leverage the opportunities available for such units [7]. Furthermore, even if the current structure of the market does not match well the strengths and weaknesses of the energy storage systems (as it was tailored to centralized thermal power generation units dominating electricity production), increasing shares of intermittent renewable generation could pressure the market into a structural change, to reflect the need of a more flexible system. Still, grid-connected Energy Storage Systems (ESS) can prove advantageous across different applications in the system, spanning from frequency control and energy arbitrage to maximizing self-consumption of renewable [8]. The biggest market in terms of volumes, in which storage units are incentivized to trade, either with or without renewable or conventional generation units, is the day-ahead (DA) market. Here, there is an opportunity for energy arbitrage, which translates into buying and selling electricity at different times, taking advantage of its price change throughout the day [2]. However, different studies showed that the profitability of various storage systems in such contexts is often not enough to repay the installation expense and guarantee a satisfying return on investment [9–11]. Because of the limited profitability observed in the day-ahead market, literature has explored alternative markets to assess the revenue potential of storage installations, including reserve markets [7,12,13]. Traditionally, ancillary services for the grid and the related bid in reserve markets, have been provided by large production units, and albeit such markets have much smaller traded volumes compared to day-ahead markets, they can offer better opportunities for energy trading and could represent interesting investment cases for storage systems [2]. This may be especially true with the increasing integration of intermittent renewable, which could induce the extension of the provision of ancillary services to smaller distributed generation and consumption units [14]. Relevant literature works include the optimization model developed in the methodological paper by Hejazi et al. [7], for the optimal operation of independent storage systems in day-ahead (DA) and reserve markets. Their work introduced a stochastic programming framework to account for the variability in the input data (wind farm production, load, etc..). The rules for the reserve market in Hejazi's study were based on an older American market, featuring a unidirectional reserve activation with no constraints. Their work showed that it was possible to ensure profitability on the investment on storage units bidding in the said market and serve as a foundational reference for the development of novel methods, especially considering the evolving landscape of reserve markets and secondary frequency regulation in particular.

Within the realm of ancillary markets, there has been a growing interest in recent years in evaluating optimal bidding strategies for battery storage systems in new types of frequency regulation services. Frequency regulation reserves are paid services acquired by the transmission system operators (TSOs) and provided by dedicated and prequalified generation units, the balancing service providers (BSPs), according to the grid rules. In the European market, the new forms of primary, secondary, and tertiary response are also referred to as FCR (Frequency Containment Reserve), aFRR (automatic Frequency Restoration Reserve), and mFRR (manual Frequency Restoration Reserve). As far as their response time: FCR has to be ready within a few seconds, then, if the frequency is still unbalanced, the TSO activates aFRR automatically after 30 s, for the following minutes after the imbalance is detected; if the grid imbalance persists after 12.5 min, the mFRR supports or replaces aFRR gradually [15]. The rules for the secondary aFRR market can vary across Europe, and each TSO acquires the necessary balancing power and energy according to its own required reaction time, price, and availability of resources. In general, units participating in aFRR have a minimum bid size of 1 MW, have to be able to deliver full power within 5 min, when committed, and have to guarantee the availability of their resources for a continuous activation period of hours, days, weeks, or even months in advance, when entering the market. Most countries in Europe have the following two distinct markets and two forms of remuneration for aFRR [15]:

- Power reserve market, involving capacity remuneration in €/MW: providers are paid for the capacity promised beforehand to the TSO, which can be activated in case of imbalances.
- Energy market, involving balancing energy remuneration in €/MWh: providers are paid for the actual energy that they absorb or inject into the grid, to compensate for the energy imbalance.

Usually, a unit can participate in the energy market and consequently receive payment, without participating in the power reserve market. On the other hand, a unit that bids on the power reserve market is required to also participate in the energy market because it has to activate the energy regulation service when required by the TSO. Usually, different prices are applied to remunerate upregulation (increasing power production to increase the frequency value) and downregulation (decreasing power production to decrease the frequency value), which can be bid separately or as a unique, symmetrical service. In case of activation, bids are called until the demand has been met. This means that an expensive bid is less likely to be accepted than a cheaper one.

Examples of the growing interest in optimal bidding strategies for battery storage systems frequency markets can be found in the following works focusing on automatic Frequency Restoration Reserve (aFRR) [16,17] and manual Frequency Restoration Reserve (mFRR) [18].

Specifically, Merten et al. [16] considered a four-hour activation period on the German/Austrian market and incorporated an aging model into the optimization, addressing this crucial aspect often overlooked in other optimization studies involving battery storage. While the paper concentrated solely on upregulation service, its insights contribute significantly to the understanding of optimal bidding strategies for batteries. In a more recent study, Nitsch et al. [17] evaluated the revenue coming from battery storage systems simultaneously participating in the day-ahead and aFRR markets, simulating scenarios for Germany in 2030 and using historical market data as an input. The linear optimization model calculated the bidding prices on the aFRR market based on the opportunity cost of not participating in the day-ahead market, providing insights for practitioners seeking to optimize battery storage system participation in these markets. The identification of the optimal size of battery energy storage systems in both DA and aFRR markets has also been of concern in [19], where the optimal size of a LiB is evaluated, using a Mixed Integer Linear Program (MILP) model that takes into account the aging mechanism of cyclic degradation, and

its optimal size was found to be at a 1 h ratio between rated capacity and rated power.

It is crucial to note that employing storage for grid applications still presents certain concerns. For example, there is uncertainty in the lifespan of storage systems within the operational context of the application because aging mechanisms, especially for batteries, are influenced by operational conditions (e.g. in the case of lithium-ion batteries there are many parameters influencing the battery aging [20]), making investments in the technology challenging due to the absence of a guaranteed business model. Furthermore, entry into frequency regulation markets is constrained to specific rules that could bring disadvantages to certain storage technologies, due to their characteristics. For instance, batteries with a fixed energy-to-power ratio like lithium-ion ones may incur high investment costs due to the extended activation period required for certain frequency regulation markets. This scenario may favor the adoption of technologies like flow batteries, characterized by lower energy-specific costs [2].

Overall, the reviewed literature suggests that exploring a relatively recent market for secondary frequency regulation, such as the aFRR, especially in combination with the day-ahead energy market, serves as an opportunity to assess the potential of specific storage technologies. The aFRR is a particularly interesting market to examine because it possesses activation times compatible with the response times of most storage technologies, specifically batteries [8], furthermore, it has a greater volume compared to the FCR market which means it offers greater investment opportunities over the latter, and the aFRR energy needs are expected to grow more than the FCR needs in the future due to the increasing penetration of renewable energy in the grid [21]. The energy storage technologies that can potentially have success in these markets include batteries with different characteristics that could be integrated into hybrid storage systems. This integration would serve to take advantage of the specific strengths offered by each technology and create synergic effects among different components while simultaneously bidding on different markets.

1.1. Hybrid storage systems

When talking about the increasing expected share of renewables on the grid, the topic of increasing temporal flexibility is also of interest. Addressing the need to decouple energy generation and consumption in both temporal and spatial dimensions, along with the imperative to tailor stationary ESS to specific time scale demands, the emergence of hybrid solutions integrating two or more storage technologies presents itself as an attractive prospect for the future. Hybrid energy storage systems (HESS) are integrated systems that leverage multiple storage technologies operating across various time scales, enabling the simultaneous provision of multiple services. The topic of hybridization of battery systems for stationary energy storage is part of the R&I activities proposed as part of the implementation plan of action 7 in the Integrated European SET Plan, which highlights the need for both efficient short and long-term storage for stationary applications to perform a combination of different services such as load-leveling, frequency regulation, provision of backup power and providing hydrogen as fuel and as feedstock for chemical industries [22]. Additionally, it suggests that employing dual systems in energy storage applications can offer benefits in terms of cost reduction, enhanced performance, increased capacity factors, and improved safety. The utilization of inexpensive batteries for storing large amounts of energy, serving the purpose of energy provision, can be combined with high-power input/output systems like Li-ion batteries, supercapacitors, or non-electrochemical systems such as flywheels [22].

These systems have the potential to address both high power peaks and high energy demands, a capability not yet found in any single energy storage technology currently available on the market at an affordable price [23]. Examples of hybrid technology combinations include pairings such as batteries and flywheels, or batteries with supercapacitors [24,25], which can be arranged in various configurations and managed through a common energy management system [26]. Yuan et al. [27] suggest through simulation results that a hybrid energy storage system comprising a flywheel and lithium battery can effectively improve grid power quality and extend its service life. Other combinations that have garnered increasing interest among researchers, can also consist of supercapacitors coupled with long-duration batteries like flow batteries, as exemplified in the EU-funded research project Hyflow [28].

1.2. Paper contribution

Expanding upon existing literature, this study presents an optimization problem that involves the optimal bidding of a wind farm paired with a hybrid battery storage system. The objective is to plan the optimal participation of this renewable system in both the day-ahead (DA) and automatic frequency restoration reserve (aFRR) markets. The assumption made in this investigation is that a storage system can concurrently submit bids in different markets that are not mutually exclusive. Within this paper, a detailed characterization of the battery storage is explored, and the solutions to the problem are compared to those obtained with a simpler battery characterization. The optimization problem is formulated and solved as a Mixed-Integer Linear Program (MILP).

The hybrid battery storage system is composed of a combination of lithium-ion batteries (LiB) and vanadium redox flow batteries (VRFB). LiBs are a technologically mature option with established costs, markets, and high efficiency, while VRFBs are novel batteries with high stability, safety, a long lifecycle, and the possibility to independently size energy and power capacities, making them suitable for long-duration energy storage [29]. These technologies are chosen in the presented case because they are considered to be among the most successful energy storage technologies for future use in energy arbitrage and secondary response applications [8], and they have been independently tested in similar applications [17,30,31], but not in a hybrid configuration, where the hybridization of high efficiency and low power cost of the LiB could find a suitable complemented pair in the low energy cost, long cyclic life of the VRFB.

In this work the following assumptions are valid: (i) the energy that the considered system has to activate when required by the TSO is always proportional to the power promised in the capacity market, (ii) the optimization program is robust to the variation of energy activated in the aFRR market, to ensure that it is always possible to provide the promised energy to the aFRR market in the worst-case scenario. Therefore, due to the presence of aFRR, a market where the product of energy activated has high uncertainty, unpredictability, and variability, the problem is solved through a robust optimization (RO) technique, in which the problem satisfies the constraints for a realistic scenario and two worst-case scenarios of maximum frequency regulation activation (maximum activation in upregulation and in downregulation). In general RO approaches can result in significantly elevated costs for the proposed solution, characterized by an excessively high "price of robustness". This is attributed to the nature of RO, which protects against every event within the specified uncertainty set, irrespective of its probability [32]. This implies that conducting a robust analysis results in conservative outcomes, offering insights into the lowest expected revenue from an investment.

The hybrid storage system is investigated as a strategic means to not only participate in the DA market but also to extend bidding capabilities to the aFRR market. This dual-market approach is motivated by the higher economic revenue opportunities offered by the aFRR market, contributing to an increased overall value of renewable sources and storage installation. Establishing a benchmark for evaluation such as comparing two separate systems or assessing the hybridization or use of two different technologies in terms of efficiency, capital, operational

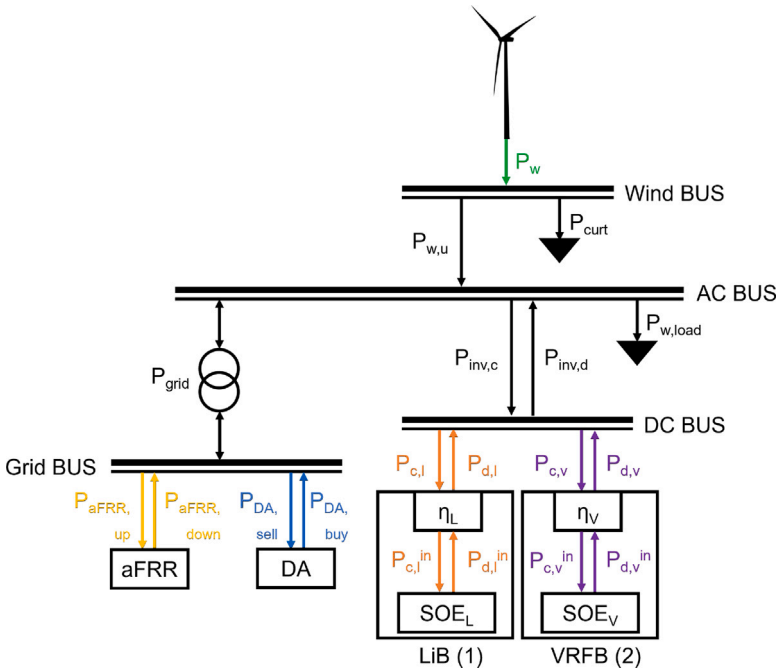


Fig. 1. Scheme of the battery-assisted grid-connected renewable system.

costs, and cycle life is integral to the analysis. To evaluate the realistic profitability gained by bidding in the aFRR market, robust optimization is used to consider uncertainty and always ensure the delivery of the promised power in the worst-case scenario while satisfying the historical demand constraints and worst-case activation constraints in upregulation and downregulation of aFRR-activated energy. The impact of robust optimization, in contrast to non-robust optimization, is investigated along with the impact of detailed storage models, which consider variable efficiencies and variable charging/discharging rates. A sensitivity analysis of the battery sizes is conducted to determine the best hybrid storage system configuration and the most profitable investment case. The economic model evaluates the ex-post life of the lithium-ion battery, after calculating its cyclic degradation as a function of the depth-of-discharge of the optimized cycles.

The main contribution of the paper is the evaluation of the impact of hybridization of lithium-ion and vanadium redox flow batteries bidding on energy and frequency regulation markets, on profitability, battery use, and the life of each storage technology.

2. Method

This section describes the mathematical model, outlining the problem statement and hypotheses, and providing a comprehensive explanation of the parameters, variables, and equations required for the formulation.

Fig. 1 presents the scheme of the modeled renewable system, containing a wind farm, feeding energy into the grid, through either DA or aFRR market, or to a hybrid battery storage system, with two batteries combined with a high degree of flexibility, and with the possibility of exchanging energy among them, in what is defined as an active parallel scheme by [33].

In this work, the unit bidding on the aFRR market adheres to the following rules: the unit promises a certain amount of capacity in the power reserve market, and determines a capacity profile for the next 24 h, with an hour resolution, to which compensation is paid in €/MW/h; furthermore, the unit has to provide energy in an amount proportional to the promised capacity, following the required activation demand, to which additional compensation is provided, in €/MWh. Both products (capacity promised and energy delivered) can be offered either for up or downregulation.

A numerical example of the bidding mechanism is the following: for any given hour of the day, if the system operator requires, for example, an activation fraction of 60% of the total allocated demand in upregulation, the unit bidding in the market will have to guarantee, for that entire hour, the delivery of upregulating energy equal to 60% of the power previously allocated in the reserve market.

Given that the focus of the problem lies in optimizing the energy dispatching on a frequency regulation market, it is crucial to address the inherent uncertainty associated with the energy activation request on this market, which cannot be predicted in advance. The inability to predict the energy request in the aFRR market could lead to an overestimation of the power promised in the capacity market. Furthermore, if a bidding unit fails to deliver the regulation energy committed in the capacity market as requested by the TSO, it could be excluded from future participation in the aFRR market.

To account for the uncertainty in this model, a robust optimization technique is employed [32]. Robust optimization (RO) is an approach that seeks to find solutions that are robust against worst-case scenarios, without assigning them a specific probability distribution. Robust optimization models aim to minimize the objective function subject to constraints that hold for all possible realizations of the uncertain parameters within a given uncertainty set (S) of scenarios, encompassing the worst-case possibilities for the aFRR market. The use of robust optimization makes the solution resilient against possible forecasting errors coming from the solution of a deterministic problem, in which a full prediction of aFRR energy request is assumed.

Fig. 2 illustrates a diagram outlining the logic behind energy exchange in DA and aFRR markets, as well as the qualitative representation of the possible energy activation scenarios determined by the uncertainty set S. It is important to note that, given the available information on renewable electricity production, electricity price, and possible activation scenarios in the day-ahead (D-1), the robust optimization ensures that the unit can always guarantee the delivery of the promised power, even in the worst-case scenarios of energy activation request. The three different scenarios in the uncertainty set are: (i) one where the energy activated in upregulation is always maximum, (ii) one where upregulation and downregulation are activated following a

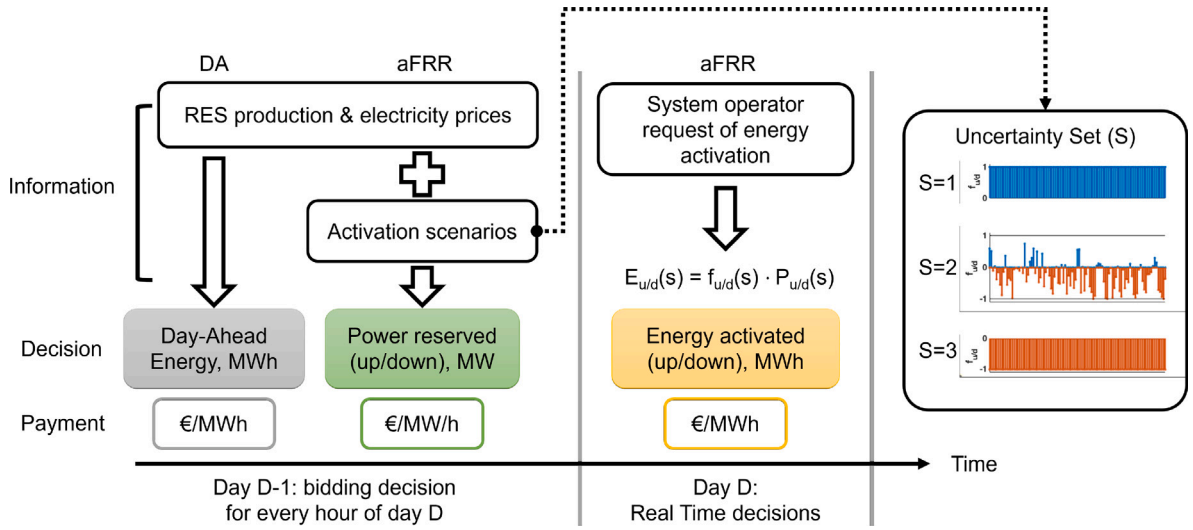


Fig. 2. Decision-making process for bidding on DA and aFRR markets. D represents the current day, in which the energy activation decisions occur on the aFRR market; D-1 represents the former day, in which the bidding decisions for DA and aFRR reserve market are taken. The uncertainty set defines the value of the fraction of up/downregulating energy required by the aFRR.

historical profile of demand, and (iii) one where the energy activated in downregulation is always maximum. Scenarios (i) and (iii) are the worst-case scenarios, where the requested energy profile consists of 100% activation fraction for the whole horizon period. The equations in the model are solved for the three different scenarios, all considered to be equally probable.

2.1. Problem statement

The optimization problem can be stated as follows. Given:

- size, efficiency, power, and state of charge limits of the batteries;
- renewable energy production hourly curve;
- price of electricity on the DA spot market;
- price of electricity up/downregulation on the aFRR market and capacity remuneration for the power reserve of up/downregulation on the aFRR market;
- different scenarios of energy activation demand on the aFRR market;

determine for each period i of the time horizon T:

- the charging and discharging power of the two batteries;
- the power exchanged with the grid;

to maximize the revenue associated with selling electricity and regulation services to the grid.

2.2. Model formulation

The optimization problem, which is solved with a one-hour resolution time, is formulated based on the hypothesis of having perfect forecasts of energy prices and renewable energy production for the next 24 h, the time horizon, and a possible set of energy activation scenarios as an input for the aFRR market. During each hour, each battery's charge and discharge phases are optimized, considering the energy prices and the energy production.

2.3. Description of parameters and variables

The optimization problem is described through known parameters and unknown decision variables, and their constraints.

The index i indicates the ith period of the day and is chosen from the set of hours in a day $T = \{1,..,24\}$. The optimization is solved for each separate day d for the entire year. Index d belongs to the set of days in a year $D = \{1,..,365\}$. The index j indicates the jth battery type and is chosen from the set $J = \{1,2\}$. The two battery types in the considered system are lithium-ion battery (LiB) and vanadium redox flow battery (VRFB). The index s indicates the scenario of energy activation on the aFRR market, and it is chosen from the uncertainty set $S = \{1,2,3\}$. The decision variables that depend on the value of the energy activated, can take different values for different scenarios and therefore depend on index s. Other decision variables do not depend on the energy activation scenario, because they are calculated on the day-ahead, and have to be valid for any value of the activated energy (i.e. the power exchanged on the day-ahead, and the power promised in the aFRR capacity market). The common abbreviations used in the model are listed in Table 1.

2.3.1. Parameters

The problem relies on a set of parameters, which can be either constant or time-varying:

- $\hat{P}_{c,max,j}$ & $\hat{P}_{d,max,j}$ [kW] - Maximum charging and discharging powers of the jth battery;
- $\hat{E}_{b,nom,j}$ [kWh] - Rated output energy of the jth battery;
- $\hat{P}_w(i,d)$ [kW] - Power generated by the wind farm, with $i \in T$ and $d \in D$;
- $\hat{P}_{w,load}(i,d)$ [kW] - Power demanded by the wind farm during idle (i.e. to power the internal electronic and hydraulic components inside the turbine), with $i \in T$ and $d \in D$;
- $\hat{P}_{g,max}$ [kW] - Maximum power that can be exchanged with the grid;
- $\hat{SoC}_{min,j}$ & $\hat{SoC}_{max,j}$ [–] - Minimum and maximum state of charge of the jth battery;
- $\hat{SoC}_{0,j}$ [–] - Initial state of charge of the jth battery at the beginning of each day;
- $\hat{c}_{DA}(i,d)$ [€/kWh] - Price of electricity on the DA market, with $i \in T$ and $d \in D$;
- $\hat{c}_{aFRR,up}(i,d)$ [€/kWh] - Price of electricity on the aFRR upregulation market, with $i \in T$ and $d \in D$;
- $\hat{c}_{aFRR,down}(i,d)$ [€/kWh] - Price of electricity on the aFRR downregulation market, with $i \in T$ and $d \in D$;
- $\hat{c}_{power}$ [€/kW/h] - Price of power reserved on the aFRR market;
- $\hat{c}^{v}_{O\&M,j}$ [€/kWh] - Variable O&M cost for the jth battery;
- $\hat{f}_{up}(i,d,s)$ [–] - Fraction of activated upregulation energy on the aFRR market;

- $\hat{f}_{down}(i,d,s)$ [–] - Fraction of activated downregulation energy on the aFRR market.

2.3.2. Variables

The problem uses the following continuous or binary decision variables:

- $P_{w,u}(i,d,s) \in \mathbb{R}^+$ [kW] - Useful power from the wind farm, with $i \in T$, $d \in D$, and $s \in S$;
- $P_{curt}(i,d,s) \in \mathbb{R}^+$ [kW] - Power curtailed from the wind source, with $i \in T$, $d \in D$ and $s \in S$;
- $P_{grid}(i,d,s) \in \mathbb{R}$ [kW] - Power exchanged with the grid, with $i \in T$, $d \in D$, and $s \in S$;
- $P_{inv}(i,d,s) \in \mathbb{R}$ [kW] - Power exchanged with the inverter, with $i \in T$, $d \in D$, and $s \in S$;
- $P_{DA}(i,d) \in \mathbb{R}$ [kW] - Power exchanged on the day-ahead market, with $i \in T$, and $d \in D$;
- $P_{aFRR,down}(i,d) \& P_{aFRR,up}(i,d) \in \mathbb{R}^+$ [kW] - Power promised in the aFRR market in downregulation or upregulation, with $i \in T$, and $d \in D$;
- $E_{aFRR,down}(i,d,s) \& E_{aFRR,up}(i,d,s) \in \mathbb{R}^+$ [kWh] - Energy delivered in the aFRR market in downregulation or upregulation, with $i \in T$, $d \in D$, and $s \in S$;
- $P_{c,j}(i,d,s) \& P_{d,j}(i,d,s) \in \mathbb{R}^+$ [kW] - Charging and discharging battery powers, external DC, with $j \in J$, $i \in T$, $d \in D$, and $s \in S$;
- $P^{in}_{c,j}(i,d,s) \& P^{in}_{d,j}(i,d,s) \in \mathbb{R}^+$ [kW] - Charging and discharging battery powers, internal DC, with $j \in J$, $i \in T$, $d \in D$, and $s \in S$;
- $\eta_{c,j}(i,d,s) \& \eta_{d,j}(i,d,s) \in \mathbb{R}$ [–] - Charging and discharging DC/AC efficiencies of the jth battery, with $j \in J$, $i \in T$, $d \in D$, and $s \in S$;
- $SoE_j(i,d,s) \in \mathbb{R}^+$ [kWh] - State of energy, with $j \in J$, $i \in T$, $d \in D$, and $s \in S$;
- $k_j(i,d,s) \in \{0,1\}$ - Binary variable which determines charging (1) or discharging (0) the battery with $j \in J$, $i \in T$, $d \in D$, and $s \in S$.

2.4. Objective function

The optimization problem involves maximizing the objective function, which comes from the sum of the revenue derived from the energy exchanged in the DA market (sold or bought) and the revenue from the aFRR market (both delivered energy and promised power contributions in upregulation or downregulation). Additionally, it accounts for subtracting the variable O&M costs for the batteries. The objective function is solved for every $d \in D$.

$$Rev(d) = Rev_{DA}(d) + Rev_{aFRR,E}(d) + Rev_{aFRR,P}(d) - \tau \cdot \sum_{j \in J} \left(\hat{c}^{v}_{O\&M,j} \cdot \sum_{i \in T} P_{d,j}(i,d,s_2) \right) \quad (1)$$

$$Rev_{DA}(d) = \tau \cdot \sum_{i \in T} P_{DA}(i,d) \cdot \hat{c}_{DA}(i,d) \quad (2)$$

$$Rev_{aFRR,E}(d) = \sum_{i \in T} E_{aFRR,up}(i,d,s_2) \cdot \hat{c}_{aFRR,up} + \sum_{i \in T} E_{aFRR,down}(i,d,s_2) \cdot \hat{c}_{aFRR,down} \quad (3)$$

$$Rev_{aFRR,P}(d) = \sum_{i \in T} \left(P_{aFRR,up}(i,d) + P_{aFRR,down}(i,d) \right) \cdot \hat{c}_{power} \quad (4)$$

τ is the time interval, equal to one hour; s_2 in Eq. (3) is the scenario containing the realistic, historical activation data for the aFRR market demand for up and downregulation.

2.5. Constraints

In this problem, due to the presence of the aFRR market, a robust optimization approach is used, and all the constraints on the energy activated and on the variables that depend on the energy activated are satisfied for the different scenarios in the set S.

2.5.1. Power constraints

The power balance on the wind farm bus, valid for every $i \in T$, $d \in D$, and $s \in S$, is given as follows:

$$\hat{P}_w(i,d) = P_{w,u}(i,d,s) + P_{curt}(i,d,s) \quad (5)$$

The power balance on the AC bus, valid for every $i \in T$, $d \in D$, and $s \in S$, is given as follows:

$$P_{w,u}(i,d,s) = P_{grid}(i,d,s) + P_{inv}(i,d,s) + \hat{P}_{w,load}(i,d) \quad (6)$$

The power balance on the grid bus, valid for every $i \in T$, $d \in D$, and $s \in S$, is given as follows:

$$P_{grid}(i,d,s) = \frac{E_{aFRR,up}(i,d,s)}{\tau} - \frac{E_{aFRR,down}(i,d,s)}{\tau} + P_{DA}(i,d) \quad (7)$$

The power balance on the DC bus, valid for every $i \in T$, $d \in D$, and $s \in S$, is given as follows:

$$P_{inv}(i,d,s) = \sum_{j \in J} \left(P_{c,j}(i,d,s) - P_{d,j}(i,d,s) \right) \quad (8)$$

Additional constraints, valid for every $i \in T$, $d \in D$, and $s \in S$, define the maximum capacity of the system's grid connection and can be formulated as follows:

$$\hat{P}_{g,max} \geq P_{aFRR,up}(i,d) + P_{DA}(i,d) \geq -\hat{P}_{g,max} \quad (9)$$

$$\hat{P}_{g,max} \geq P_{aFRR,down}(i,d) - P_{DA}(i,d) \geq -\hat{P}_{g,max} \quad (10)$$

$$\hat{P}_{g,max} \geq \frac{E_{aFRR,up}(i,d,s)}{\tau} + P_{DA}(i,d) \geq -\hat{P}_{g,max} \quad (11)$$

$$\hat{P}_{g,max} \geq \frac{E_{aFRR,down}(i,d,s)}{\tau} - P_{DA}(i,d) \geq -\hat{P}_{g,max} \quad (12)$$

The maximum capacity constraints on the promised aFRR capacity (Eqs. (9) and (10)) are valid for every $i \in T$, and $d \in D$; the maximum capacity constraints on the activated aFRR energy (Eqs. (11) and (12)) are valid for every $i \in T$, $d \in D$, and $s \in S$. The maximum capacity is due to a limited physical connection with the grid, which is assumed to be equal in both directions.

2.5.2. aFRR market constraints

The activated energy for the aFRR market, which has to be delivered by the system, is always proportional to the committed power, through a fraction $f_{up/down}$, for each $i \in T$, $d \in D$, and $s \in S$:

$$\frac{E_{aFRR,up}(i,d,s)}{\tau} = \hat{f}_{up}(i,d,s) \cdot P_{aFRR,up}(i,d) \quad (13)$$

$$\frac{E_{aFRR,down}(i,d,s)}{\tau} = \hat{f}_{down}(i,d,s) \cdot P_{aFRR,down}(i,d) \quad (14)$$

At a certain time, the required regulation need is either in the up or down direction, and the fractions $\hat{f}_{up}$ and $\hat{f}_{down}$ are never simultaneously active for a certain scenario.

2.5.3. Storage constraints

To guarantee that each battery respects its charging and discharging maximum power limits, and prevent simultaneous charge and discharge, the following constraints are introduced, for every $j \in J$, $i \in T$, $d \in D$, and $s \in S$:

$$P_{c,j}(i,d,s) \leq k_j(i,d,s) \cdot \hat{P}_{c,max,j} \quad (15)$$

$$P_{d,j}(i,d,s) \leq (1 - k_j(i,d,s)) \cdot \hat{P}_{d,max,j} \quad (16)$$

Additionally, the storage has to follow constraints on the maximum capacity and minimum amount of energy that can be stored every time, which correspond to constraints on the state of energy (SoE). The state of energy is related to the battery charging and discharging powers:

$$SoE_j(i,d,s) = \hat{SoE}_{0,j} + \tau \cdot \sum_{x=1}^{i} \left(P_{c,j}(x,d,s) \cdot \eta_{c,j}(x,d,s) - \frac{P_{d,j}(x,d,s)}{\eta_{d,j}(x,d,s)} \right) \quad (17)$$

$$\hat{SoC}_{min,j} \cdot \frac{E_{b,nom,j}}{\bar{\eta}_{d,j}} \leq SoE_j(i,d,s) \leq \hat{SoC}_{max,j} \cdot \frac{E_{b,nom,j}}{\bar{\eta}_{d,j}} \tag{18}$$

where $\bar{\eta}_{d,j}$ is the average discharge efficiency of the jth battery. When considering a simple battery characterization, the efficiencies are assumed to be constant in time, therefore parameters of the problem, and equal to the average values of the charge/discharge variable efficiencies. The problem is therefore linear and it is solved with a Mixed Integer Linear Program (MILP) without any additional required step. In the case of variable efficiencies, instead, the problem becomes a MINLP and needs further linearization techniques to be solved as MILP. The passages of the linearization are explained in the supporting information document (SI1.1). We assume that the battery efficiencies are already inclusive of the inverter and other power converter losses.

Finally, the state of energy of the storage at the last time step of the day must be consistent with the initial energy state at the beginning of each day, ensuring continuity in the state of energy from one day to the next:

$$SoE_j(i,d,s)\Big|_{i=24} = \hat{SoE}_{0,j} \tag{19}$$

2.6. Economic modeling

The results of the different HESS designs and market configurations in terms of investment profitability are compared using the following indexes: Discounted Payback Period (DPBP) and Net Present Value (NPV).

Assuming to have constant annual net cash flows (CF_i), the Net Present Value at the end of the investment, after N_y years, is calculated as follows:

$$NPV = -C_{0,tot} + \sum_{i=1}^{N_y} \frac{CF_i}{(1+r_{dy})^i} \qquad [€] \tag{20}$$

where $C_{0,tot}$ is the total discounted cost of the storage system and r_{dy} is the annual discount rate.

The Discounted Payback Period measures the time it takes for the investment to recover the initial costs, and equals the time at which the NPV becomes positive.

$C_{0,tot}$ is calculated as the sum over the set J of initial ($C_{0,j}$) and discounted replacement costs ($C_{0,j}^{rep}$) of each jth battery.

$$C_{0,tot} = \sum_{j \in J} \left(C_{0,j} + C_{0,j}^{rep} \right) \qquad [€] \tag{21}$$

$$C_{0,j} = E_{b,nom,j} \cdot \hat{c}_{E,j} + P_{d,max,j} \cdot \hat{c}_{P,j} \qquad [€] \tag{22}$$

$$C_{0,j}^{rep} = \left(E_{b,nom,j} \cdot \hat{c}_{E,j}^{rep} + P_{d,max,j} \cdot \hat{c}_{P,j}^{rep} \right) \cdot f_j^{rep} \qquad [€] \tag{23}$$

Replacement costs, which are divided into energy-specific and power-specific, are actualized using the replacement factor f_j^{rep}, evaluated using the formulation from [34]:

$$f_j^{rep} = \sum_{k=1}^{n_j^{rep}} \frac{1}{(1+r_{dy})^{k \cdot t_j^{rep}}} \qquad [-] \tag{24}$$

The number of replacements of a certain part of the storage system (n_j^{rep}), throughout the investment period (N_y), depends on the replacement time interval (t_j^{rep}) as follows:

$$n_j^{rep} = ceil\left(\frac{N_y}{t_j^{rep}} \right) - 1; \qquad [-] \tag{25}$$

The replacement time interval is considered to be fixed for the vanadium battery, and equal to 10 years, the time after which membrane and electrodes are replaced into the battery stack [35]. In this case, no energy-related replacement is considered, because the electrolyte of the vanadium battery undergoes periodic restoration (i.e.: electrolyte rebalancing and servicing [36]) to restore the capacity reversibly lost during cycling. For the VRFB, the cost of capacity restoring is considered to be included in the O&M cost terms.

On the other hand, the replacement time interval for the lithium-ion battery corresponds to its life (L_{LiB}), at the end of which a replacement of the battery pack occurs, and the specific replacement cost for this battery equals the initial energy specific cost of the pack. The life of the LiB is the minimum value between its cyclic life and its calendar life, equal to 10 years [35]. The cyclic life is evaluated using a function of its cyclic degradation, strongly influenced by the depth-of-discharge (DOD) during cycling, as explained in the supporting information document (SI1.2).

Ultimately, it is assumed that the annual net cash flows (CF_i) of the investment remain constant each year, as the periodic replacements are incorporated into the initial total cost of the storage and no periodic subsidies are considered for the investment.

$$CF = \Delta Rev - C_{O\&M,tot}^{f} \qquad [€/\text{year}] \tag{26}$$

The net cash flow is calculated by subtracting the total fixed operational costs of the system from the net revenue generated by the investment (ΔRev).

$$\Delta Rev = Rev - Rev_{ref} \qquad [€/\text{year}] \tag{27}$$

Rev equals the annual revenue from the system, calculated as the sum of the values of the objective functions through the year; Rev_{ref} is the annual revenue evaluated for the reference case, i.e. the case without a storage system.

The annual operational costs of the HESS are divided into (i) variable O&M costs, embedded inside the objective function of the optimization, (ii) fixed O&M costs, which are calculated as a fraction of the capital cost:

$$C_{O\&M,tot}^{f} = \sum_{j \in J} C_{O\&M,j}^{f} = \sum_{j \in J} \hat{c}_{O\&M,j}^{f} \cdot C_{0,j} \qquad [€/\text{year}] \tag{28}$$

3. Case study

This section introduces the case study to which the optimization model is applied. It contains the model input data and offers insights into the reasoning behind the choices made for the input parameters.

The optimization problem, in which the objective is to maximize the stacked revenue coming from the energy wholesale arbitrage on the DA market and the energy reservation and activation on the aFRR market, is solved for a Danish case study based on historical data. An energy storage system is assumed to be installed in a portion (10 MW out of 40 MW) of the existing onshore wind power plant of Lem Kær [37], equipped with MW-scale Vestas turbines, located in Central Denmark, to maximize the profit from the energy exchange, according to the fluctuations in day-ahead energy market price and the aFRR regulation needs. The available energy market data is for 2021. The wind farm data is given by Vestas with a resolution of 10 min, and it is hourly aggregated. The annual average wind farm capacity factor is 38%. Table 2 contains the description and the values of all the constant parameters necessary to solve the optimization model.

Since the regulations of the secondary reserve market vary across different countries, this work refers to generic aFRR market rules, which may apply with little variations to most of the existing systems. In particular, a unit can freely bid on both the DA and aFRR market, and specifically, a unit bidding on the aFRR market is remunerated both for the capacity bid as power reserve and for the actual activated energy requested by the system operator. Both products (capacity promised and energy delivered) can be offered either for up or downregulation.

For specific characteristics and prices, the paper looks at the anticipated aFRR rules in the future Danish market, expected to be in place from 2024. This case study is chosen due to Denmark's notable position featuring one of the highest proportions of variable renewable penetration in the electric grid [3]. Specifically, Denmark holds the

Table 2
Symbol, name, unit of measurement, value, and source of the parameters in the optimization model.

Symbol	Name	Unit	Value	Source
$\hat{P}_{W,nom}$	Nominal wind farm size	MW	10	–
$\hat{P}_{d,max,1}$	Nominal LiB discharging power	MW	0.001–5	–
$\hat{P}_{d,max,2}$	Nominal VRFB discharging power	MW	0.001–2	–
$\hat{P}_{g,max}$	Maximum grid power	MW	10	–
$\eta_{RT,1}$	Average LiB roundtrip efficiency	%	83	[38]
$\eta_{RT,2}$	Average VRFB efficiency	%	65	[39]
$\hat{SoC}_{min,1}$&$\hat{SoC}_{max,1}$	Minimum and maximum LiB SoC	%	20–100	–
$\hat{SoC}_{min,2}$&$\hat{SoC}_{max,2}$	Minimum and maximum VRFB SoC	%	0–100	–
$\hat{SoC}_{0,j}$	Initial SoC	%	50	–
$t_{d,1}$	Nominal discharging LiB time	h	1	–
$t_{d,2}$	Nominal discharging VRFB time	h	1–6	–
$\hat{c}_{power}$	Price of power on the aFRR market	€/kW/h	0.0204	–
$\hat{c}_{E,1}$	Initial LiB energy specific cost	€/kWh	321	[40][a]
$\hat{c}_{P,1}$	Initial LiB power-specific cost	€/kW	304	[40][a]
$\hat{c}_{E,2}$	Initial VRFB energy-specific cost	€/kWh	186	[41]
$\hat{c}_{P,2}$	Initial VRFB power-specific cost	€/kW	1080	[41]
$\hat{c}^{f}_{O\&M,j}$	Fixed storage O&M costs	%/year	0.43	[35][a]
$\hat{c}^{v}_{O\&M,j}$	Variable storage O&M costs	€/MWh	0.45	[35][a]
$\hat{c}^{rep}_{E,1}$	Energy-specific replacement cost of LiB	€/kWh	321	[40][a]
$\hat{c}^{rep}_{P,1}$	Power-specific replacement cost of LiB	€/kW	0	
$\hat{c}^{rep}_{E,2}$	Energy-specific replacement cost of VRFB	€/kWh	0	[41]
$\hat{c}^{rep}_{P,2}$	Power-specific replacement cost of VRFB	€/kW	350	[41]
N_y	Investment time	years	20	–
t^{rep}_{2}	Replacement time of VRFB	years	10	[35][a]
r_{dy}	Annual discount rate	%	8	[8]

[a] Average annual energy exchange rates for years 2020 and 2023 were used to convert costs from $ to € [42].

highest share of wind energy in both total primary energy consumption and electricity among all IEA countries [43].

The Danish market for regulation services is divided into 2 bidding zones: the western part of Denmark, DK1, connected to the European Synchronous Area, and the eastern part, connected to the Nordic Synchronous Area, DK2 [44]. The Danish TSO, Energinet, ensures that a total capacity of ±100 MW for aFRR is always available in the DK1 area, the reference area for this case study [44]. The acquisition of the capacity has been done during a weekly auction since 2022, changing from the previous monthly auctions, due to a lack of competition in the market, to lower the market risk for suppliers of the reserve, attracting more balancing service providers [21]. The capacity has been procured as a symmetrical product (equal amount of up and downregulation provided) until now. From 2024, aFRR capacity and aFRR energy will be procured separately: the capacity will be procured through the Nordic capacity market (expected to have a total volume of about 300 MW) with separate purchases of aFRR up and downregulation, while aFRR energy will be procured through the European aFRR energy market, PICASSO, also with separate up and downregulation [21]. Since the scope of the work is to investigate the potential revenue of the future aFRR market, the reference market has the features expected for the 2024 Nordic aFRR market, except for the data for prices and activation scheduling, which are taken from 2021. The characteristics of the Nordic aFRR market are listed in Table 3.

Because in 2022 not all the auctions were satisfied, and therefore there are missing values in the value of the product on the capacity market, the year 2021 was taken instead as a reference for DA spot market energy prices in DK1, and aFRR activation prices (equal to DA spot prices ±100 DKK/MWh [47], with DKK to €conversion according to the average 2021 exchange rate of 0.1345 €/DKK [42]). Historical aFRR activation energy schedule is taken from [48]. The average pricing for aFRR capacity in 2021, determined by the average prices on a pay-as-bid compensation approach, amounted to 303 DKK/MW/h [47]. Anticipating an asymmetrical product in the future Nordic market, we project that the cost of either upregulation or downregulation capacity on the aFRR market will be approximately half of the average price of the symmetrical product, estimating it to be around 151 DKK/MW/h (20.4 €/MW/h) for this case study.

Table 3
Expected future characteristics of the Nordic aFRR market.
Source: [45,46]

Parameter	Value
Bid frequency	Daily
Market size	100 MW
Full activation time	15 min
Symmetrical product	No
Bid structure	Hourly
Minimum bid amount	1 MW
Maximum bid size	50 MW
Bid size resolution	100 kW
Capacity compensation	Marginal pricing (pay-as-cleared)
Energy compensation	DK1 spot ± 100 DKK/MWh

4. Results and discussion

This section presents and discusses the most relevant technical and economic results associated with the optimal participation of wind farms and HESS in the day-ahead and automatic frequency restoration reserve markets. The results contain the evaluation of the impact of the hybrid battery system installation on a wind farm, with a sensitivity analysis of the storage size. The impact of the storage installation and the hybridization are evaluated in terms of profitability, battery use, and life of the storage. Furthermore, the impacts of robust programming and detailed battery models are evaluated.

The optimization problem is solved using a computer with the following specifics: CPU Intel(R) Core(TM) i7-6800K with a 3.40 GHz clock speed, with 6 physical cores and 12 logical processors. The

Table 4
Gross revenue (M€/year) coming from a wind plant ($P_{n,wind}$ = 10 MW) for different investment cases without storage or with different storage systems (BESS) configurations of a baseline size (LiB: 1 MW/1 MWh, VRFB: 500 kW/1 MWh), bidding on the day-ahead (DA) market independently or collectively with the aFRR market.

Revenue [M€/year]	W/O storage	Non-hybrid BESS	HESS
DA	3.06	3.07	3.09
DA + aFRR	4.24	4.33	4.42

problem with constant battery efficiencies, comprised of 144 binary variables and 936 continuous variables for each daily horizon period, is solved inside the Yalmip [49] environment for Matlab, using the Gurobi [50] solver, with an average time of 0.7 s per day. The problem with variable battery efficiencies, comprised of 1008 binary variables and 1800 continuous variables for each horizon period, is solved with an average time of 9.6 s per day. The MIP Gap of the problem was set to 0.01%.

4.1. Impact of hybridization of storage on different markets

The results of different investment cases are evaluated for the following baseline storage sizes:

- Lithium-ion battery: 1 MW/1 MWh
- Vanadium battery: 500 kW/1 MWh

This combination is among the most successful cases of hybrid storage system sizes, among the ones tested in the sensitivity analysis, as shown in Section 4.2.

If not specified otherwise, the batteries are assumed to have variable efficiencies and variable maximum charging and discharging rates.

Table 4 shows the results in terms of gross annual revenue for different cases of wind farm installations. The system without storage is comprised of the sole renewable power plant, with a nominal size of 10 MW. In contrast, the system with battery storage (BESS) can take two different configurations: either in a hybrid or non-hybrid form. In the hybrid storage system (HESS), the batteries are integrated and work collaboratively, allowing for the exchange of energy between them to optimize the overall storage and grid exchange of renewable energy. On the other hand, in the non-hybrid BESS, the two batteries are independently managed without exchanging energy between them. In this case, each technology operates autonomously, and it is optimized separately from the other, without any direct interaction or energy transfer, as if there were two distinct renewable systems with two separate batteries. Both batteries, in this scenario, can bid on the aFRR market, since they do not have any relevant ramping-up limitation in regards to the secondary regulation needs. The hybrid and non-hybrid scenarios are assumed to have the same components, which are managed differently, therefore they also have the same capital cost.

Results demonstrate a consistent increase in annual revenue from the renewable system when bidding also on the aFRR market, compared to a system exclusively bidding on the DA market. This results in a relative revenue boost ranging from 39% to 56%, regardless of whether storage (hybrid or non-hybrid) is present. Participating in the aFRR market enables the unit to earn additional revenue, because the aFRR market, in contrast to the DA market, provides remuneration not only for the energy exchanged on the market but also for the promised power on the reserve market, even when the energy is not provided. So, when participating in the aFRR market, part of the energy that would be bid on the DA market is instead allocated on the aFRR reserve market and exchanged on the aFRR energy market when requested by the system operator.

The amount of energy procured in the aFRR market in 2021 was equal to 98 GWh in upregulation and 216 GWh in downregulation [48]. The energy exchanged on the DA market in DK1 for the same year was about 22 TWh [48]. Therefore, the volume of the DA market in Denmark is more than 100 times bigger than the volume of the aFRR market. This means that, even though participation in the aFRR market offers more opportunities than participation in the sole DA market since the former market has a smaller volume than the latter, it might saturate more quickly and have more competitors, resulting in a reduction in the remuneration of allocated power. In fact, for the future Nordic aFRR market, from 2024 onwards, the aFRR capacity will be paid according to market rules, with marginal pricing (pay-as-cleared) instead of pay-as-bid. Greater unit participation will, therefore, reduce the paid pricing for all participants, and units bidding energy with a higher price will be excluded from the market.

Table 5
Discounted payback period (DPBP) coming from the installation of different storage systems configurations on a wind plant ($P_{n,wind}$ = 10 MW). The storage system (BESS) has a baseline size (LiB: 1 MW/1 MWh, VRFB: 500 kW/1 MWh) and it is either in hybrid or non-hybrid configuration, bidding on the day-ahead (DA) market independently or collectively with the aFRR market.

Investment scenario	DPBP [years]		Non-hybrid BESS	HESS
	Initial case	Upgrade		
A	DA	DA	>100	>100
B	DA	DA + aFRR	2	2
C	DA + aFRR	DA + aFRR	>100	19

It is important to note that the theoretical scenario where the renewable system bids concurrently on the DA and aFRR markets without any storage, is allowed in this context to distinguish the impact of the storage from the impact of bidding on the aFRR market as an addition to bidding on the DA market. This case finds a correspondence to reality, because the Danish market permits bidding on the aFRR market with a wind turbine, without the presence of storage, provided that specific bidding rules are followed [51].

The three investment scenarios are further illustrated in Table 5. For all investments, A, B, and C the investment upgrade consists of the installation of the storage, either hybrid or non-hybrid. In the case of investment B, other than the installation of the storage, there is a further upgrade, from bidding on the sole DA market to bidding on DA + aFRR markets. As confirmed by the existing literature, exclusively bidding on the DA market (investment scenario A) results in a financially unfeasible investment case, as the generated net profits are insufficient to cover the costs of the storage over the operational lifespan of the system, with negative Net Present Values, and Discounted Payback Periods (DPBP) of more than 100 years. Under the assumption of a reference case where the system bids only on the DA without any storage, the investment associated with the storage installation and simultaneous bidding on the aFRR market (investment scenario B) is very promising, with DPBP values of about 2 years, regardless of the type of storage system installed. If a storage system is evaluated against a reference case already comprising of DA + aFRR bidding (investment scenario C), the investment case is not promising, resulting in a DPBP of 19 years for the hybrid system, and a DPBP greater than 100 years for the non-hybrid system. For this case, there is a substantial difference in the DPBP of the two storage system configurations, which means that even if hybridization contributes to a small revenue increase (see Table 4), this is enough to modify the results of the investment. This holds, even though the price of the storage is still too high to guarantee a satisfactory return on investment at the current market conditions, assuming conservative conditions as in the present work.

The effects of hybridization on the number of optimal yearly cycles of the batteries are analyzed for different cases of relative battery sizes:

- when the LiB is greater than the VRFB, in terms of rated energy, the number of cycles of the LiB tends to decrease (−6% on average), while the number of cycles of the VRFB tends to increase (+38% on average);

Table 6
Lithium-ion battery life (years) with different storage systems (BESS) configurations of a baseline size (LiB: 1 MW/1 MWh, VRFB: 500 kW/1 MWh), while bidding on the day-ahead (DA) market independently or collectively with the aFRR market.

Life [years]	Non-hybrid BESS	HESS
DA	7.5	7.5
DA + aFRR	9.1	10

- when the LiB is smaller than the VRFB, in terms of rated energy, the number of cycles of the LiB does not experience significant change, while the number of cycles of the VRFB increases slightly (+10% on average)

In general, with the hybrid combination of the two batteries, part of the energy load appears to be transferred from the LiB to the VRFB, which works with more energy, increasing its average cycles per year compared to the non-hybrid case, regardless of the hybrid battery system size. This does not impact the overall life of the VRFB, as its cyclic life is not sensitive to depth-of-discharge or number of cycles. The capacity reversibly lost during the cycling of the VRFB can be restored during periodic O&M operations, at a variable cost, proportional to the energy stored inside the battery, without additional capital expenditure.

The impact of hybridization is also evaluated in terms of the storage operating life, as the lithium-ion battery undergoes faster degradation as the depth-of-discharge (DOD) increases. Table 6 shows the results for the baseline case, in terms of LiB life, calculated using the degradation model described in the supporting information document (SI1.2), for different system configurations. In the DA case, the lifespan of the lithium-ion battery remains consistent, irrespective of the system configuration, be it non-hybrid or hybrid. On the other hand, in the DA + aFRR case, the life of the lithium-ion battery increases by about 10% when the battery is collocated in the hybrid system, with respect to the case in the non-hybrid system, in the baseline case. For the different investigated storage sizes (see Section 4.2), generally the effect of the hybridization is positive on the lithium-ion battery, which increases its operating life span up to 47% compared to the case without hybridization.

The effect of the markets on the LiB life is various. Regardless of the storage system size, the number of cycles and cycling strategy of the LiB on the DA market yields similar results, and the life of the battery stays within the range of 7.1–7.8 years. On the other hand, with the presence of the aFRR market, depending on the size of the storage system, the battery life stays in the range of 6.7–10 years, with much greater variability compared to the sole DA case. In general, bigger batteries tend to have a smaller cyclic lifetime, because they undergo more frequent cycling, at higher DOD. In general, the life of the LiB tends to increase if the battery is in the hybrid configuration because the energy is shared between the LiB and the VRFB: this helps reduce the number of cycles at high depth-of-discharge, which benefits the life of the lithium-ion battery. This effect is much more evident in the DA + aFRR market, where the effects of hybridization on the LiB life are greater than in the sole DA case. This is caused by the fact that, due to the characteristics of the DA market, the batteries are subject to fewer limitations compared to the DA + aFRR case, therefore they store greater amounts of energy, even without the hybridization. In the sole DA market, batteries have higher degrees of freedom, either in hybrid or non-hybrid configuration, therefore the benefits of the hybrid configuration either on the revenue or on the lithium-ion battery life are less evident in this case, compared to the DA + aFRR market.

4.2. Sensitivity to storage size

For the sensitivity analysis, the size of each battery (lithium-ion or vanadium RFB) is changed independently from the other. The investment case consists of the upgrade obtained from the installation of a storage system to bid on the aFRR market, starting from an initial system with no storage, which bids only on the DA market (investment scenario B, see Table 5).

The results of different investment cases have been evaluated for the following storage sizes:

- Lithium-ion battery: power range 0.001–5 MW; discharging time: 1 h;
- Vanadium battery: power range 0.001–2 MW; discharging time: 1–6 h;

The performance of the lithium-ion battery chosen for this case has been evaluated for a discharging C-rate up to 5C [38]. Here, due to the high power-specific costs of the battery (e.g. made of balance of plant costs, power converter costs, etc, … [40]), and due to the specific needs of the case, the lithium-ion battery is limited to a nominal discharging rate of 1C and therefore it has a nominal discharging time equal to 1 h. The VRFB can have an independently sized energy and power module and takes a nominal discharging time between 1 and 6 h. VRFBs can reach bigger energy-to-power ratios (e.g. 10 h [24]), but for this case, these values have not been investigated, as above 6 h of discharge the investment cases become less interesting as the payback time increases notably due to the high investment costs.

The results for the sensitivity analysis on different storage sizes for the HESS are shown in terms of Net Present Value (NPV) and Discounted Payback Period (DPBP), for investment scenario B, in Figs. 3 and 4. Investment scenario B compares a system with hybrid storage, bidding on both DA and aFRR markets, to an original system that does not have storage and bids only on the DA market. Results are shown for different values of lithium-ion battery power, vanadium RFB power, and vanadium RFB discharge time (t_d).

Looking at the Net Present Value (Fig. 3), the best outcome in terms of the storage system size is given by the 1 MW lithium-ion battery, installed with the smallest tested VRFB in terms of power (1 kW). The best LiB among the ones tested, with a size of 1 MW/1 MWh, has a rated power equal to 10% of the peak wind farm power (10 MW), and a rated energy equal to only 1.1% of the average amount of energy daily generated by the wind farm (91 MWh). Since the recommended VRFB size to maximize the NPV is negligible compared to the optimal LiB size, results hint that the best system in economic terms is composed solely of the lithium-ion battery, without any VRFB installed. However, very similar results in terms of NPV are shown for the 500 kW/1 MWh VRFB. In general, if a VRFB of any size were to be installed in the hybrid system, with the preference for smaller capacities, its best discharge time would be around 2 h. Installing a VRFB by itself will always be less convenient, in terms of NPV, than adding a small (1–2 MW) LiB in hybrid configuration alongside it.

Fig. 3 shows an outlier. The NPV of a hybrid system composed of a 5 MW LiB and a 0.5 MW/2 h VRFB is higher than expected, diverging from the trend observed in other configurations, and showing a non-convex effect in the results. This happens because, despite the similar revenue and the higher capital cost associated with this combination compared to the case with a smaller VRFB, the synergic effect between the batteries in this scenario significantly extends the lifespan of the LiB. Consequently, this effect is impactful enough to lower the total lifetime system cost, leading to a bigger NPV, which surpasses the NPV of the case with a smaller VRFB for the same LiB size.

All the cases for investment scenario B show fairly low discounted payback periods (see Fig. 4), in the range of 2–7 years for the investigated storage sizes.

Although the hybrid storage system installation and participation in DA and aFRR markets can generate consistent profits based on the assumptions explored, these markets tend to favor smaller-capacity batteries with discharge times of 1–2 h. This observation aligns with previous findings as documented in [19]. Consequently, VRFBs, which have higher power-specific costs (1080 €/kW) compared to lithium-

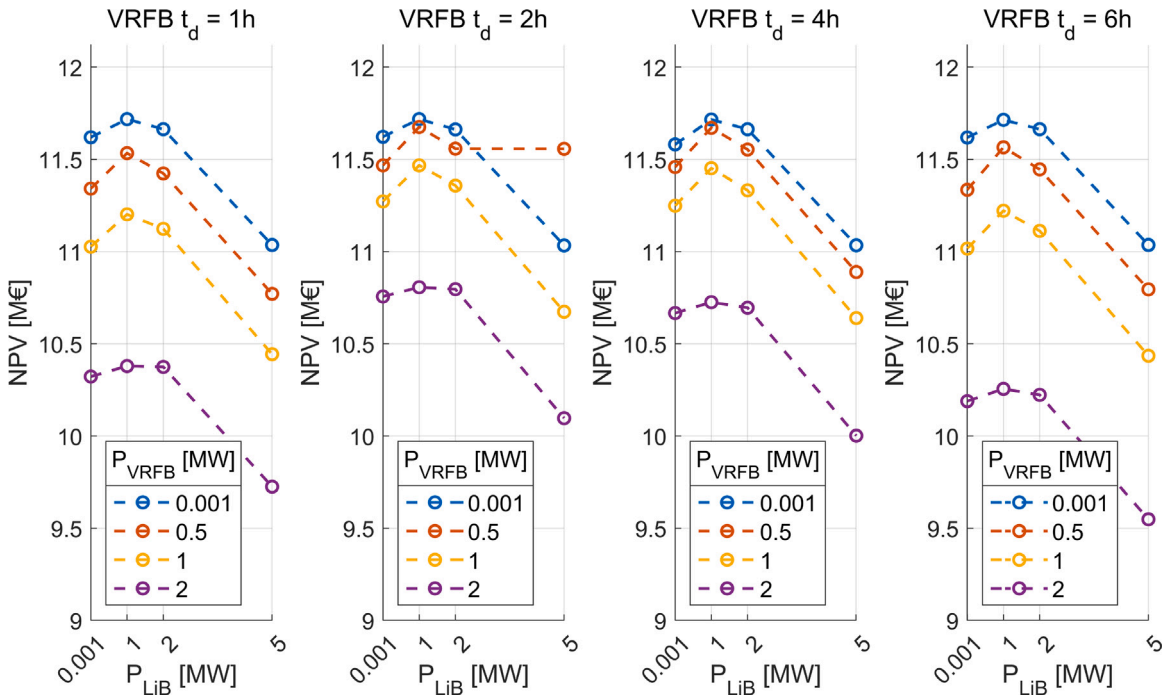


Fig. 3. Net Present Value (NPV) coming from a wind plant ($P_{n,wind}$ = 10 MW) bidding on DA and aFRR market after the installation of a HESS for different storage sizes, for investment scenario B.

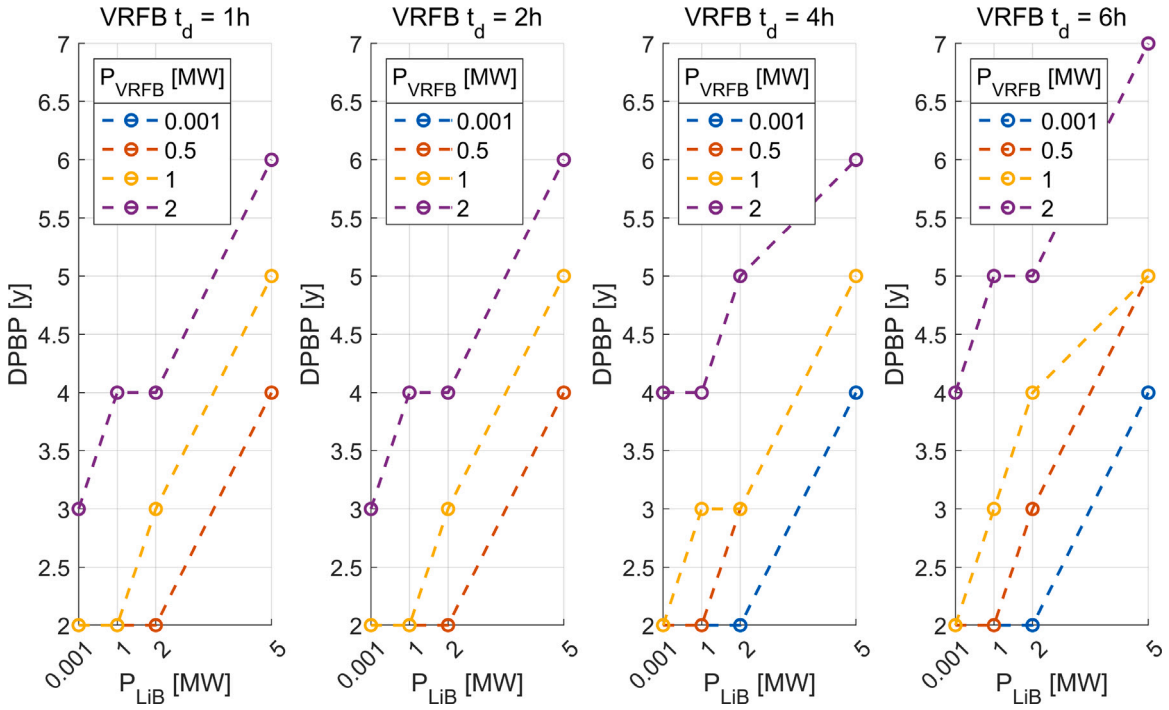


Fig. 4. Discounted Payback Period (DPBP) coming from a wind plant ($P_{n,wind}$ = 10 MW) bidding on DA and aFRR market after the installation of a HESS for different storage sizes, for investment scenario B.

ion batteries (304 €/kW), are less economically viable in systems requiring low energy-to-power ratios. As such, they are not well-suited for synchronization with the timing characteristics of the wind source and the trading markets analyzed in this study. This is unless there is a significant shift in market structure and regulatory requirements in the future that would economically and technically incentivize the use of long-duration energy storage, above 4–6 h, or on the other hand, unless the VRFB will reduce its power-specific costs.

4.3. Impact of detailed battery models

The results shown in the previous sections considered detailed battery models, comprising variable efficiencies depending on charging and discharging powers, and variable maximum power rates depending on the state of charge. In this section, those results are compared with the ones obtained with a simple model, where efficiencies and maximum power rates are considered constant, highlighting the difference

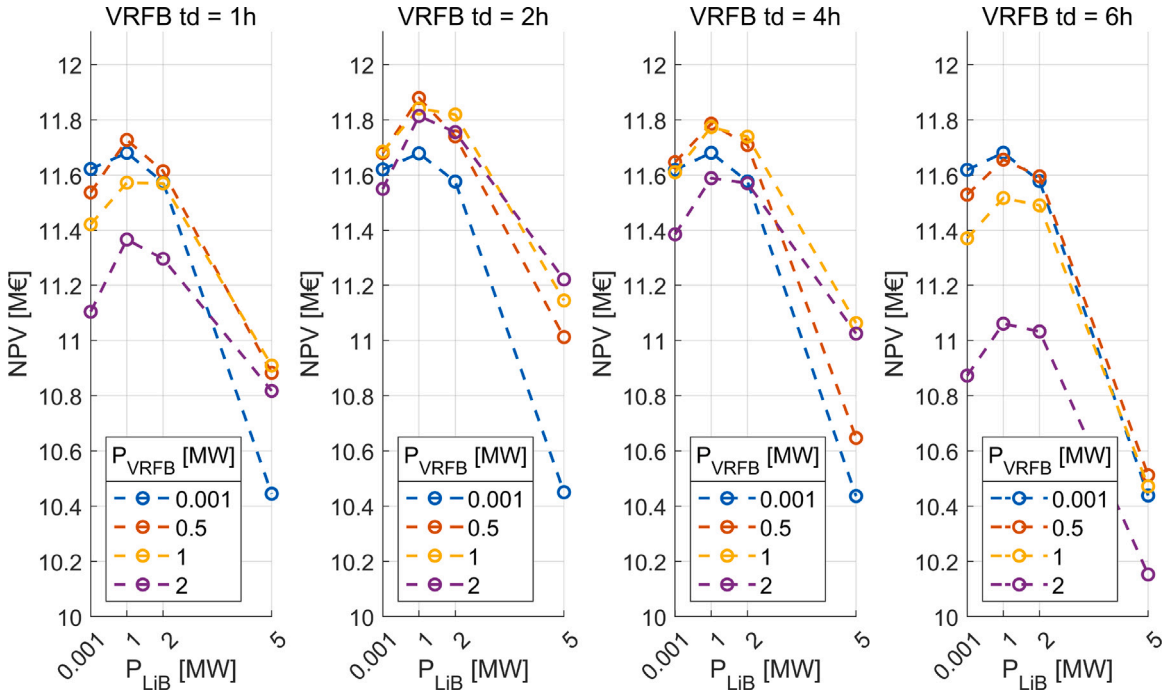


Fig. 5. Net Present Value (NPV) coming from a wind plant ($P_{n,wind}$ = 10 MW) bidding on DA and aFRR market after the installation of a HESS for different storage sizes, for investment scenario B, assuming a simple battery model.

that the models yield in terms of revenue and optimal hybrid storage system size.

Fig. 5 shows the results in terms of net present value, as in Fig. 3, for the investment scenario B and for different hybrid storage system sizes, obtained using a formulation with simple battery models. Using a simple battery model, the NPV stays more or less the same for investment scenario B, ranging from 0.95 to 1.1 times the NPV values obtained from the detailed battery model calculations. The difference becomes more visible in the case of investment scenario C, where the investment is influenced by the sole effect of the storage system implementation, and where the NPV calculated using the simple model ranges from −9.6 to 9.4 times the one calculated with detailed models, depending on the HESS size.

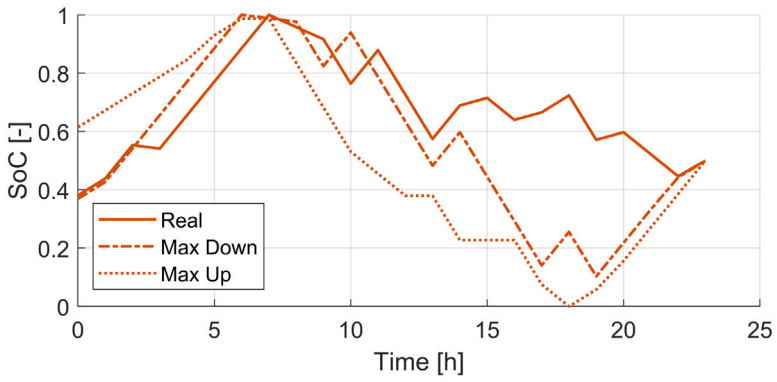
Furthermore, in both investment scenarios B and C, the best battery size changes depending on the model employed. The utilization of a constant efficiency model tends to overestimate the revenue generated by the HESS, consequently influencing the optimal storage size necessary to maximize the NPV. As illustrated in Fig. 5, choosing a VRFB ranging from 500 kW to 2 MW can often yield greater benefits compared to having the smallest VRFB (1 kW), differently from what is suggested by results with the detailed model. However, the best size for lithium-ion batteries (1 MW/1 MWh) and the best discharge time for VRFB (2 h) remain consistent, indicating their relatively low sensitivity to battery characterization. This suggests that these parameters exhibit a high level of robustness, confirming the significance of the findings.

On the other hand, for the presented case, assessing the detailed models' impact on the projected NPV proves challenging. The variations in the modeling of efficiencies and maximum power rates can significantly influence the overall investment viability prediction, potentially resulting in either an increase or decrease in revenue value. This outcome is highly sensible to the evaluated storage system's size, highlighting the difficulties involved in evaluating the system's economic viability.

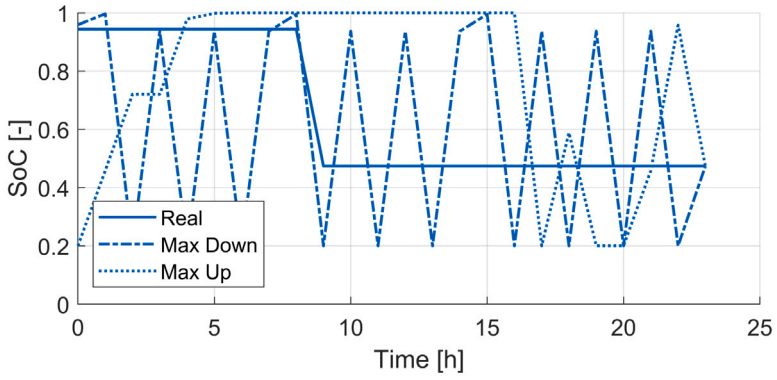
4.4. Impact of robustness

In the mathematical formulation proposed, robust optimization implies the solution of the set of constraints for three scenarios. One scenario represents the realistic activation of energy in the aFRR market, while the two other scenarios represent the worst case, the total activation of energy for aFRR, either in upregulation or downregulation. By optimizing for the worst-case scenario within these uncertainty sets, any power promised is guaranteed to be delivered even under the most unfavorable conditions. The solution is considered to be conservative of the possible revenue coming from bidding in the aFRR market. To mathematically model robust optimization, both aFRR and storage constraints have to be satisfied for the three uncertainty scenarios. The problem is going to have virtually three State of charge (SoC) variables for each timestep, as can be seen in Fig. 6, which depicts an example of the optimal battery state of charge throughout a day, for different scenarios in the robust optimization, for both lithium-ion and vanadium battery. The realistic SoC value is going to be the effective SoC of each battery, while the other two values correspond to virtual states which could occur in case of 100% aFRR energy activation either in up or downregulation (worst-case scenarios). In these cases, the battery has deeper and more numerous cycles than in the real case.

One way to evaluate the effects of robust optimization is to compare the results of this formulation to the results of a non-robust case, where the problem is solved not by taking into account any worst-case scenario, but by assuming a perfect forecast for energy activation request on the aFRR market. The annual revenue calculated using the non-robust formulation is equal to 2.2–2.3 times the revenue calculated using the robust formulation, assuming HESS sizes in the range described in Section 4.2, while the NPV is equal to 5.6–6.5 times the one calculated with the non-robust formulation, for scenario B. These results show that a non-robust formulation leads to a non-conservative preliminary evaluation, potentially greatly overestimating the actual revenue from bidding into the aFRR market.



(a) LiB



(b) VRFB

Fig. 6. Example of optimal LiB and VRFB SOC (state of charge) for different scenarios of the robust optimization: realistic, worst-case of maximum up-regulation activation, worst-case of maximum down-regulation activation.

The lowered revenue of the robust optimization, as hinted, comes from not being able to predict the actual energy activation needs on the aFRR markets, and from the need to lower the promised capacity to the value that can be completely fulfilled. This, in turn, means not utilizing the resources at their full potential because cautious scheduling must be adopted for battery capacity management, which generally leads to a lower number of cycles performed by the storage system. In particular for the baseline case (see Section 4.1), the number of average charge/discharge cycles per day equals 1.1 for the LiB and 1.2 for the VRFB with the robust optimization, on the other hand, for the non-robust optimization, the number of cycles equals respectively 4.3 (LiB) and 2.6 (VRFB) cycles per day. Other cases, with different HESS sizes, have similar results. Therefore, at least for the baseline case, a non-robust formulation may overestimate the optimal number of cycles of each battery by about 116%–290% compared to a robust formulation.

In essence, results show that the price of robustness is high when participating in the aFRR market, significantly diminishing anticipated profits compared to a non-robust case. In theory, this line of work estimates the lowest end of profit expected when bidding on the aFRR market under the presented assumptions. In practice, units intending to participate in the aFRR market, despite the inability to forecast future aFRR energy requirements, must make relatively cautious decisions to ensure they can reliably deliver the necessary activation energy, albeit with some level of uncertainty. The high price of robustness in an unpredictable market underscores the need for a tailored stochastic approach to prepare bidders for uncertain future scenarios, without being so conservative to overly reduce expected profits. Additionally, the impact of penalties on the aFRR market – not considered here – for the bidders unable to consistently deliver the previously promised power in the capacity market, will also affect profit and influence the decision-making of the participating units.

5. Conclusions

This study introduces an optimization problem focusing on the optimal bidding strategy for a hybrid battery storage system coupled with a wind plant in both the day-ahead (DA) and automatic frequency restoration reserves (aFRR) markets. The renewable system coupled participates in energy trading by offering renewable energy on the market, leveraging energy arbitrage opportunities in the DA market. Additionally, it allocates a portion of its available capacity to the secondary regulation market, the aFRR, on which the energy request is activated by the system operator within a few minutes following a frequency imbalance occurrence, to maintain grid stability.

The research explores the addition of a hybrid energy storage system (HESS) composed of lithium-ion (LiB) and vanadium redox flow batteries (VRFB) to a pre-existing onshore wind plant located in Denmark. Both hybrid (HESS) and non-hybrid battery energy storage systems configurations are considered, allowing for a collaborative or independent operation of the batteries.

The optimization problem is formulated as a MILP, with the assumption of having a perfect forecast of renewable energy production and electricity market price for a horizon time of 24 h. A robust optimization (RO) approach is employed to address uncertainties for the energy activation required by the aFRR service. In this problem, batteries are modeled using detailed models, assuming variable efficiencies, variable maximum power rates, and different replacement conditions, depending on the calendar and cyclic life of the battery technology.

A sensitivity analysis on battery sizes reveals the impact of varying configurations on profitability, battery use, and operating life span. The investigated wind farm has a peak power of 10 MW, and alongside the plant, different configurations of HESS sizes were tested, with

lithium-ion batteries in the range of 0.001–5 MW of power, and 1 h of discharging time, and vanadium batteries in the range of 0.001–2 MW of power, and 1–6 h of discharging time.

The key contributions and findings are summarized as follows:

- As expected, the annual revenue shows a substantial increase (39%–56%) when bidding in both DA and aFRR markets, compared to exclusive participation in the DA market. The additional participation in the aFRR market can potentially impact the overall valorization of renewable sources and storage installations on the market.
- Hybridization of the technologies positively influences the investment, and a hybrid storage system yields a lower payback time compared to a non-hybrid system. Nevertheless, the price of the storage is still too high to guarantee a return on investment in the investigated wholesale markets. Furthermore, the installation of a storage system proves to have a less positive impact on the profits than expanding the market participation of the renewable system from the DA to the aFRR market.
- Hybridization has in general, a positive effect on the cyclic life of the lithium-ion batteries, with a relative increase of up to 47% compared to the case without hybridization.
- The sensitivity analysis on the storage size calculates the net present value (NPV) and the discounted payback period (DPBP) of an investment consisting of integrating the HESS into a system bidding either DA or both DA and aFRR markets. The best HESS proves to be composed substantially of a 1 MW/1 MWh LiB, yielding a Net Present Value (NPV) of about 11.7 M€ after 20 years of investment, and a projected simple payback period of 2 years, on an original case where the system bid only on the DA market.
- Despite the increase in computational burden, modeling storage with detailed models provides different results compared to the case of simple models. Detailed battery models can influence the NPV, which ranges from −9.6 to 9.4 times the value calculated with simple models, depending on the size of the storage. This highlights the challenges in accurately predicting the investment viability using simple battery models.
- The robust optimization approach ensures conservative evaluations, considering worst-case scenarios. Using a non-robust formulation, the annual revenue from the system with hybrid storage is equal to 2.2–2.3 times the revenue calculated using the robust formulation, depending on the size of the HESS.

CRediT authorship contribution statement

Diana Cremoncini: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Guido Francesco Frate:** Writing – review & editing, Supervision. **Aldo Bischi:** Writing – review & editing, Supervision. **Tim Tørnes Pedersen:** Writing – review & editing, Software, Methodology, Conceptualization. **Gorm Bruun Andresen:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Anders Bentien:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Lorenzo Ferrari:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Anders Bentien reports a relationship with Visblue that includes: co-founder, board membership. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.est.2024.112309.

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